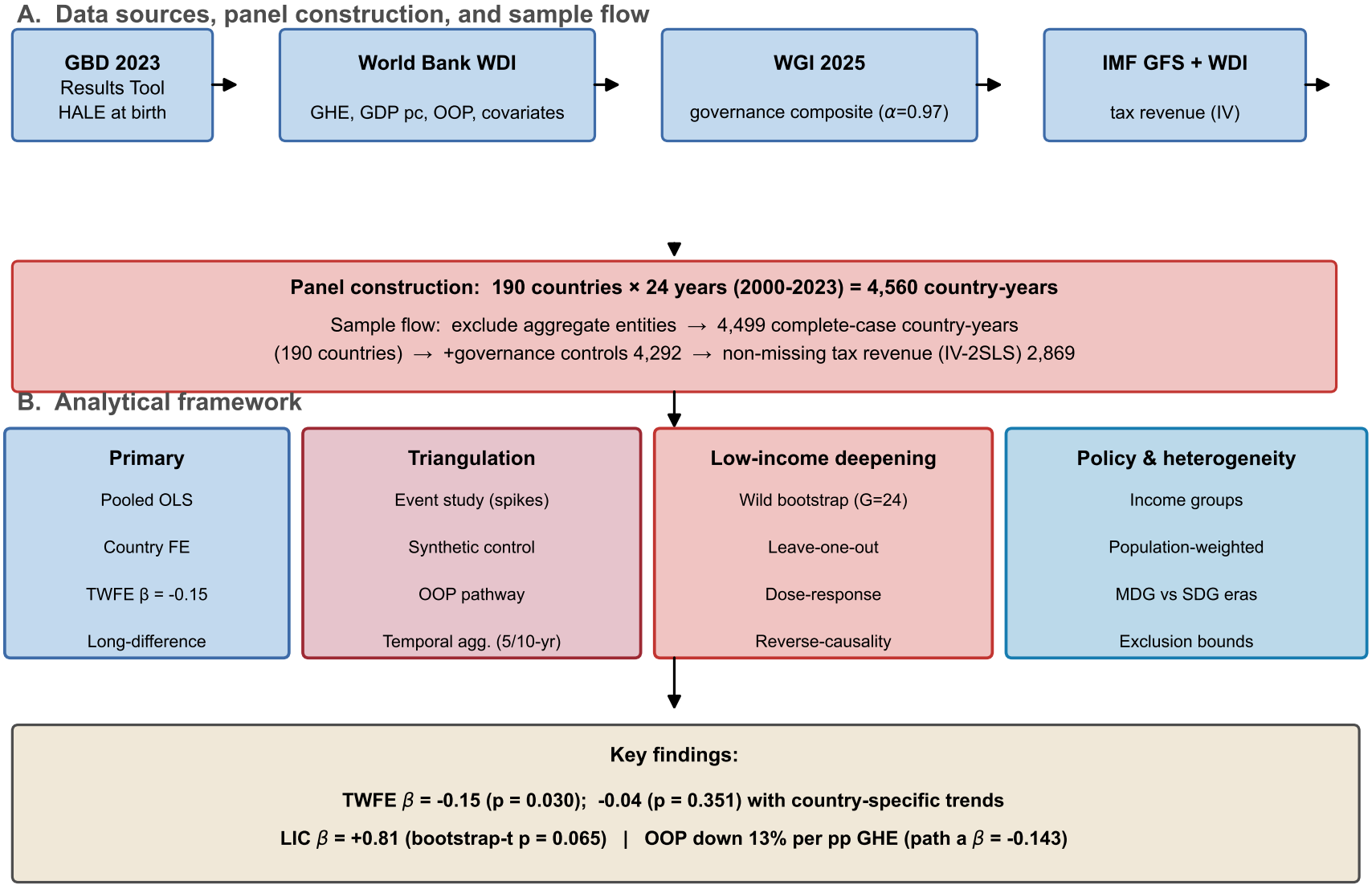
A. Data sources, panel construction, and sample flow
GBD 2023
Results Tool
HALE at birth
World Bank WDI
GHE, GDP pc, OOP, covariates
WGI 2025
governance composite (α=0.97)
IMF GFS + WDI
tax revenue (IV)
Panel construction: 190 countries × 24 years (2000-2023) = 4,560 country-years
Sample flow: exclude aggregate entities → 4,499 complete-case country-years
(190 countries) → +governance controls 4,292 → non-missing tax revenue (IV-2SLS) 2,869
B. Analytical framework
Primary
Pooled OLS
Country FE
TWFE β = -0.15
Long-difference
Triangulation
Event study (spikes)
Synthetic control
OOP pathway
Temporal agg. (5/10-yr)
Low-income deepening
Wild bootstrap (G=24)
Leave-one-out
Dose-response
Reverse-causality
Policy & heterogeneity
Income groups
Population-weighted
MDG vs SDG eras
Exclusion bounds
Key findings:
TWFE β = -0.15 (p = 0.030); -0.04 (p = 0.351) with country-specific trends
LIC β = +0.81 (bootstrap-t p = 0.065) | OOP down 13% per pp GHE (path a β = -0.143)